Maternal Diet Alters the Sensory Qualities of Human Milk and the Nursling's Behavior

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ABSTRACT. Although the majority of human infants are breast-fed for the first few months of life, there is a paucity of information regarding the sensory qualities of human milk and how these qualities are affected by maternal diet. The present study investigated the effects of garlic ingestion by the mother on the odor of her breast milk and the suckling behavior of her infant. Evaluation of the milk samples by a sensory panel revealed garlic ingestion significantly and consistently increased the perceived intensity of the milk odor; this increase in odor intensity was not apparent 1 hour after ingestion, peaked in strength 2 hours after ingestion, and decreased thereafter. That the nursling detected these changes in mother's milk is suggested by the finding that infants were attached to the breast for longer periods of time and sucked more when the milk smelled like garlic. There was a tendency for infants to ingest more milk as well; the lack of a significant effect may be due to the inherent limitations on the total amount of milk available to the infant. *Pediatrics* 1991;88:737–744; *breast-feeding, infant behavior, lactation, maternal diet, olfaction, sucking, breast milk, garlic.*

During the past decade there has been a resurgence in breast-feeding such that the majority of human infants in the United States and other Western countries are being breast-fed for the first several months of life.[1,2] But although human milk of often the sole food source for the nursling, there is a paucity of information regarding its sensory qualities and whether such qualities are affected by maternal diet.[3] And if diet does affect the taste, odor, and consequently flavor of breast milk, what are the immediate and long-term effects of maternal dietary choices for the nursling?

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While these issues remain unresolved for humans, some relevant nonhuman, animal data are available. As early as the 18th century, it was reported that cow's milk can acquire a wide variety of flavors from the female's feed.[4] That the sensory experiences associated with early milk feedings affect later food preferences has also been demonstrated. Weanling animals, for example, prefer the diet ingested by their mother during lactation[5-7] and are more likely to accept unfamiliar foods if they experienced a wide range of flavors during suckling.[8]

Although little is known about the sensory experiences associated with breast-feeding in humans, we do know the human infant is sensitive to a wide variety of gustatory and olfactory stimuli.[9,10] Of special interest is the saliency of breast odors in the dyadic interaction between mother and infant. Breast-fed infants, for example, spend more time orienting toward a breast pad previously worn by their lactating mother than one worn by an unfamiliar lactating woman.[11,12] That these infants are responding, at least in part, to volatile compounds in their mother's milk is suggested by the finding that the incidence of mouth and head movements increases when infants are presented with flasks containing mother's milk but not when presented with empty flasks.[13,14] Thus, infants may be using olfactory cues emanating from the breast region, along with visual, tactile, and thermal cues, to locate the nipple.[15]

The first aim of the present study was to determine whether the odor of human milk is altered because certain flavors in the mother's diet are transmitted to her milk. That human milk may be altered is strongly suggested by the following findings. First, dietary components contribute to odors emanating from the hands, breath, urine, and fecal material of humans.[16] Second, complex molecules ingested by a lactating female (eg, drugs) readily pass into her milk.[17] Finally, the milk of other mammals acquires flavors from the female's feed.[18] We chose garlic as the initial dietary component to investigate because it is a natural food product frequently consumed by humans and because the literature on dairy cows suggests that major sulfur-containing volatiles found in garlic are transmitted to milk, resulting in garlic-flavored milk.[4,18]

The second aim of the present study was to determine whether changes, if any, in the sensory qualities of human milk affect the behavior of the nursling during breast-feeding. For each feed, we monitored such variables as time attached to the nipple, amount of milk consumed, and the patterning of sucking. Because of individual variation in infant feeding patterns and because the duration of feeds and the amount of milk consumed varies throughout the day,[19,20] a within-subjects design was used.

METHOD

Subjects

Eight women (seven multiparous, one primiparous) who were exclusively breast-feeding a 3- to 4-month-old infant were recruited from the University of Pennsylvania community and local La Leche groups. The mothers' ages ranged from 25 to 40 years (median = 30 years), and their infants' ages ranged from 90 to 121 days (median = 97 days). All mothers and infants were in excellent health. One additional mother-infant dyad began but did not complete the study and another did not comply with the schedule and thus was disqualified.

Informed consent was obtained from each mother prior to entry into the study, which had been approved by the Committee on Studies Involving Human Subjects at the University of Pennsylvania. To assess how often the mother consumed sulfurous foods during pregnancy and lactation, a food frequency questionnaire listing 20 sulfur-containing foods (eg, garlic, onions, broccoli) was completed by each subject.[21] This revealed that none of the mothers were regular consumers of garlic but all ate it and other sulfur-containing foods occasionally.

Mothers were also instructed to record how often and from which breast they nursed their infants dring the 3 days prior to each testing day. These nursing logs revealed that the average number of breasts the infants suckled from during the 4 hours in which testing occurred (approximately 9:30 AM to 12:30 PM was 4.6 (SE = 0.3). This information enabled us to determine whether the frequency of nursing observed during the two testing periods was characteristic for each mother-infant dyad.

Procedure

Each mother-infant dyad was tested on 2 days separated by 1 week. On each testing day, mothers were instructed not to wear any perfumes or to use scented soaps and deodorants in an attempt to limit the odor overtones worn by the mother. To minimize the amount of sulfurous compounds in the mother's milk at the beginning of each testing day, subjects were instructed not to eat foods containing sulfur during the 3 days prior to each day of testing. They were given a list of foods (eg, garlic, onion, asparagus) and spices (eg, garlic powder, onion powder) to avoid. To encourage compliance, they were asked to record (in terms of household measures) all foods and beverages consumed during the 3-day period. On each testing day, we monitored the dietary intake of each subject and provided them with a variety of foods to eat, in all of which sulfur was absent or very low in concentration. Mothers consumed lunch at approximately the same time on each day of testing.

The mother arrived at Monell at approximately 9:30 AM, having last fed her infant at the same time on each of the two testing days. Within a half hour of arrival, she expressed, usually from one breast only, a baseline sample of approximately 20 mL of foremilk by using an electric breast pump (Medela, Inc, Crystal Lake, IL). This was repeated approximately every hour (±15 minutes) thereafter for 4 hours. The variation in collection time was allowed so that nursing time would be disrupted as little as possible. (One mother, however, could not express the third sampling, so only four samples were obtained from her on each day of testing.) Each sample of milk was immediately placed on ice in a sterilized glass container. After the second expression, hereafter referred to as time 0, mothers ingested either placebo capsules or capsules containing a total of 1.5 g of garlic extract (General Nutrition Center, Pittsburgh, PA). Thus, for all mothers except one, five samples were collected at hourly intervals of 1 hour before (hereafter referred to as baseline) and 0, 1, 2, and 3 hours after ingesting the capsules. Four of the mothers ingested the placebo during the first day of testing; for the remaining four, the order was reversed. No effect of order was observed for any of the variables investigated.

Evaluation of the Nursling's Behavior

Each infant was breast-fed at the frequency customary for each nursing pair. Each mother chose which breast the baby suckled from first and whether the second breast was proffered. From this, we determined the number of breasts the infant suckled from during each 4-hour testing session. Because the interval between detaching from one breast and attaching to the other is sometimes very short, we used the method of Drewett and colleagues[22] to determine the number of feeds per testing session. A feed was defined as the period of attachment to the breast such that if the interval between detaching from one breast and reattaching to the other was less than 5 minutes, then these adjacent breast-feeds were treated as components of a single feed.

Before being put to the breast and at the end of each feed, the baby was weighed (without a change in clothing) on a Mettler PM 15 top-loading balance to determine the weight of the milk consumed.[23,24] This balance, accurate to 1.0 g, was programmed to account for movements made during weighing. The volume of milk consumed by the infant was then estimated by dividing the change in the infant's weight by the specific gravity of mature human milk; the latter is 1.031.

During each feed, the baby's face was videotaped. These videotapes were later analyzed by using a Commodore 64 Personal Computer with a specially written program. One key was pressed each time the baby was seen to suck and another when the infant detached from the nipple.[25] Output generated from this program included such variables as the number of sucks per feed and the total time attached to the nipple. Two individuals, one of whom was "blind" to the conditions, scored the tapes; the intercoder agreement was very high (time attached: $r(34) = .99$; number of sucks per feed: $r(34) = .97$). All summary statistics are expressed as mean ± SEM.

Because the ingestion of spicy foods, especially those containing garlic, by a nursing mother is often associated with colic in her breast-fed infant, we asked each mother whether she noticed any change (eg, abdominal cramps, colic) in her infant the day following ingestion of the garlic extract and placebo. None of the mothers reported any noticeable change in her infant the day following ingestion of the garlic or placebo.

Sensory Evaluation of Milk

To determine whether adults could discriminate a difference in the odor of the breast milk as a function of garlic or placebo ingestion, a sensory panel evaluated the samples of breast milk. The panel consisted of 11 individuals (six women, five men) with normal olfactory functioning. To qualify, each panelist had to exhibit normal thresholds for pyridine, an odorant few people fail to detect.[26] Tasting of the milk was not attempted because the human immunodeficiency virus has been isolated in human milk[27] and some evidence suggests it can be transmitted from mother to infant via infected breast milk.[28] Thus, the panel evaluated the milk samples on the basis of odor alone.

Shortly after the last sample was collected, the five samples of milk were brought to room temperature, and 10 mL of each was individually placed in a 250-mL polypropylene, plastic squeeze bottle with a plastic flip-up cap; bottles and caps were boiled before use to minimize their odor. At least seven panelists evaluated each set of milk samples. All possible pairs of samples (5 samples yield 10 possible pairs) in each set were presented twice, one pair at a time, to each panelist individually. Using a forced-choice procedure, the panelist was asked to indicate which of the pair smelled "stronger" or "more like garlic." Panelists could sniff as often and as long as they wished and were unaware of the type of capsule the mother had ingested that day.

The null hypothesis tested was that the perceived odor of the five samples did not vary significantly across the collection period regardless of whether garlic or placebo was ingested. For each test day (garlic or placebo), a summary score was computed for each panelist by determining the number of times each sample was chosen; this number could vary between 0 and 8. (If all five samples smelled the same, for example, the average expected score for each sample would be 4 or 50%.) After these scores were cast into a two-way table having k columns (number of samples) and N rows (number of panelists), they were ranked from 1 to k. The Friedman two-way analysis of variance by ranks test was then performed to determine significance.[29] Two separate analyses were conducted for each mother, one for the sensory evaluation data obtained on the day she ingested the placebo and another for that obtained on the day she ingested the garlic.

RESULTS

Sensory Evaluation of Milk

Evaluation of the milk samples by the sensory panel revealed that for every mother tested, there was no perceived change in the odor of the milk across the collection period on the day the placebo was ingested (see Fig 1). Panelists chose the baseline milk samples and samples obtained 0, 1, 2, and 3 hours after ingestion of the placebo, on average, 51.1 ± 3.3%, 49.0 ± 2.6%, 50.4 ± 4.0%, 50.6 ± 3.3%, and 48.7 ± 3.7% of the time, respectively. A value of 50% was expected if there was no difference in the odor of the samples and response by panelists was random.

In contrast, a statistically significant variation in the perceived odor of the milk was observed for each mother on the day garlic was ingested (Friedman test, all P values < .05). That is, the sensory panel chose the baseline milk samples and samples obtained at 0 and 1 hour postingestion of the garlic 30.7 ± 3.1%, 35.2 ± 3.5%, and 48.0 ± 4.4% of the time, respectively. This percentage increased to an average of 78.6% of the time (SE = 3.6) when panelists evaluated the samples 2 hours postingestion of the garlic and then decreased to an average of 57.3% (SE = 6.9) when the samples obtained 3 hours postingestion of the garlic were evaluated (see Fig 1). It should be noted, however, that for three of the mothers, panelists continued to choose the samples obtained 3 hours after garlic ingestion at frequencies similar to that observed for 2 hours (78.0 ± 1.3%, n = 3). Thus, a change in odor was observed in the milk of each of the mothers 2 hours after garlic ingestion and in the milk of three of the mothers again at 3 hours postingestion of the garlic.

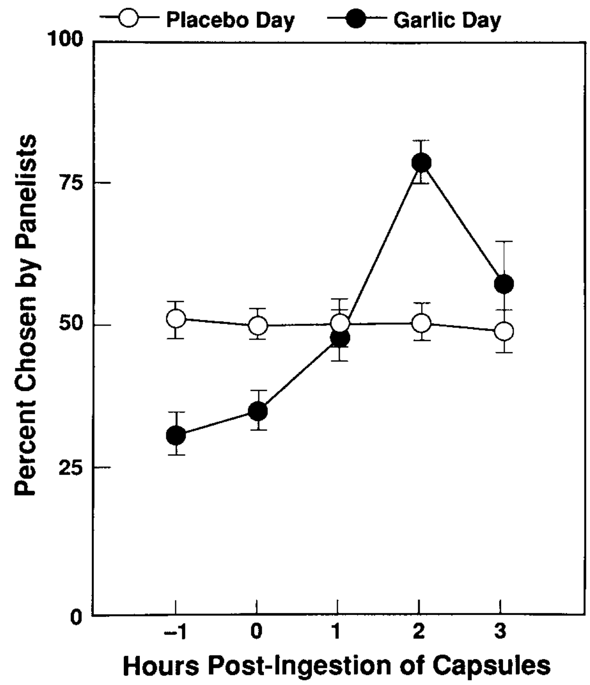

Fig 1. Percentage of time panelists chose milk samples taken 0, 1, 2, and 3 hours postingestion of garlic (closed circles) and placebo (open circles). Using a forced-choice paradigm, panelists were presented individually with each set of milk samples and asked to indicate which of the pair smelled "stronger" or "more like garlic." If there was no difference among the samples, a 50% frequency would be expected. Values are expressed as mean ± SEM.

Evaluation of the Nursling's Behavior

There was no significant difference in the number of breasts the infants suckled from during each of the 4-hour testing sessions (placebo vs garlic: 5.1 ± 0.6 vs 4.2 ± 0.2; t = 1.8, not significant [NS]); this was similar to that reported by the mothers in their nursing logs and thus enabled us to conclude that the frequency of nursing observed during the two testing periods was characteristic for each mother-infant dyad. Likewise, the number of feeds was similar during the two testing sessions (placebo vs garlic: 3.5 ± 0.5 vs 3.1 ± 0.4 times; paired $t[7]$ = 2.05, NS, see Table).

Although there was no significant difference in the number of times the baby fed, infants spent significantly more time attached to the nipple during the 4-hour testing period in which mothers ingested the garlic (32.8 ± 6.6 minutes) compared with the testing session in which the placebo was ingested (27.4 ± 5.2 minutes; paired $t[7]$ = 3.03, P < .02, see Table). The analysis of variance revealed a significant interaction between the day of testing (placebo or garlic) and the timing of the feeds (before or after the odors first appeared in the milk, $F[1,7]$ = 18.75; P < .005). That there was no difference in the time spent suckling during those feeds that occurred prior to or within 1.5 hours of capsule ingestion (placebo vs garlic: 15.0 ± 2.8 vs 14.0 ± 3.4 minutes; paired $t[7]$ = 1.01, NS) indicates that the infants were at similar baseline levels on the 2 days of testing. In contrast, a significant difference was

TABLE. Effect of Garlic Ingestion by the Mother on the Behavior of the Infant During Nursing

Behavior of Nursling	Type of Capsules Ingested by Mother	
	Placebo	Garlic
Time attached to the nipple, min		
Prior to or within 1.5 h after ingestion	15.0 ± 2.8	14.0 ± 3.4
1.5–3 h after ingestion	12.4 ± 2.6	18.8 ± 3.6*
Total 4-h testing period	27.4 ± 5.2	32.8 ± 6.6*
Amount of breast milk consumed, mL		
Prior to or within 1.5 h after ingestion	101.2 ± 13.1	111.3 ± 15.7
1.5–3 h after ingestion	69.0 ± 5.4	92.8 ± 17.3
Total 4-h testing period	170.2 ± 13.7	204.1 ± 31.1
No. of feeds		
Prior to or within 1.5 h after ingestion	1.9 ± 0.3	1.5 ± 0.3
1.5–3 h after ingestion	1.6 ± 0.3	1.6 ± 0.2
Total 4-h testing period	3.5 ± 0.5	3.1 ± 0.4

* P < .05.

found in those feeds occurring 1.5 to 3 hours after ingestion of the capsules. That is, infants were attached to the breast for longer periods of time when the milk smelled stronger or more like garlic (placebo vs garlic: 12.4 ± 2.6 vs 18.8 ± 3.6 minutes; paired $t[7] = 4.33$, $P < .005$). This increase in the time spent suckling was observed in all but one of the infants tested (see Fig 2).

Further analyses of the video records also revealed a significant increase in the number of sucks observed in those feeds occurring when the milk smelled like garlic (placebo vs garlic: 637.6 ± 164.5 vs 946 ± 208.9; paired $t[6] = 4.08$, $P = .007$; the video records of one infant did not provide enough clarity to determine sucking). Thus, infants were not only attached to the breast for a longer period of time but they sucked more also.

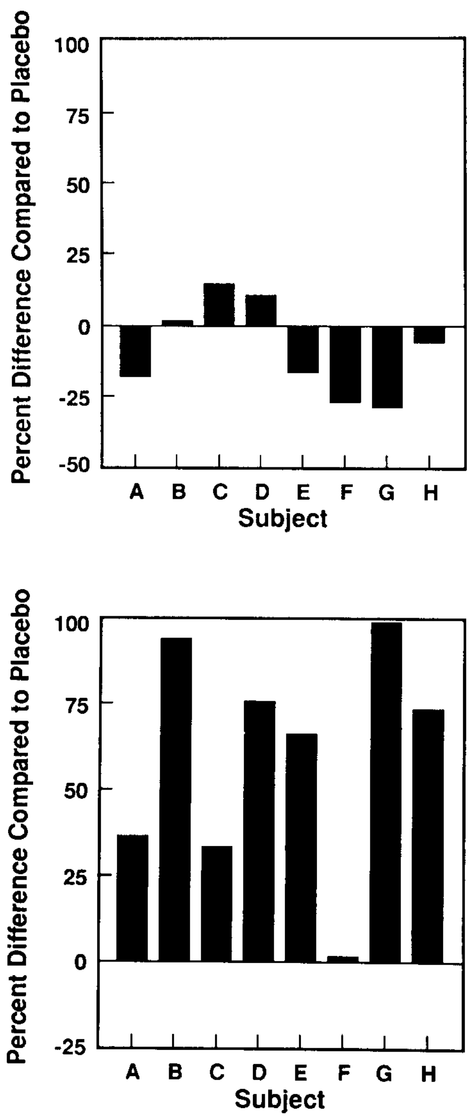

Fig 2. Percent difference in time each infant was attached to the nipple on the day their mother ingested garlic when compared with time attached on the day their mother ingested placebo. Top graph: feeds occurring prior to or within 1.5 hours after ingestion of the capsules. Bottom graph: feeds occurring 1.5 to 3 hours postingestion of capsules. See text for statistical analyses.

Despite this increase in suckling, there was no significant difference in the amount of breast milk consumed during the two 4-hour testing periods (placebo vs garlic: 190.2 ± 13.7 vs 204.1 ± 31.1 mL; $t(6) = 1.54$, $P = .18$; weights could not be obtained for one infant). Nor was there a significant difference in amount consumed during those feeds that occurred 1.5 to 3 hours after capsule ingestion (placebo vs garlic: 69.0 ± 5.4 vs 92.8 ± 17.3 mL; paired $t[6] = 1.49$, $P = .19$; see Table). It should be noted, however, that three of the infants consumed 50% to 142% more of the "garlic-flavored" milk compared with the amount consumed 1.5 to 3 hours after the mother ingested the placebo; the remaining four infants consumed similar amounts on both days of testing (±25% difference). For two of the latter infants, there may have been little milk left in their mother's breast; these mothers reported that their breasts felt empty and it took them longer to express milk during the subsequent collection. Whether these infants would have consumed more if the milk supply were larger is not known.

DISCUSSION

Effects of Maternal Diet on Sensory Qualities of Human Milk

The present study shows that the diet of a lactating woman alters the sensory qualities of her milk. Garlic ingestion significantly and consistently increased the perceived intensity of milk odor as judged by adult panelists. This increase in odor intensity was not apparent 1 hour after ingesting garlic, peaked in strength 2 hours after ingestion, and decreased thereafter. Such a time course is consistent with that observed for cows ingesting highly odorous silage.[30]

Although some of the panelists perceived the odor as garlic-like, this was not the case for all of them. Previous research has revealed that volatile sulfur-containing compounds contribute to the flavor and odor of cow's milk following the ingestion of garlic weeds and other odorous feeds[31]; this most likely is what was detected by the panelists. However, direct chemical analysis combined with sensory analysis is needed to substantiate this. Panelists also reported that the strength of odor was relatively weak. Because the human nose can generally detect very low levels of sulfur,[32] the concentration in the milk presumably was low.

Garlic was selected for testing because of its known ability to produce off-flavors in the milk of dairy cows and for its well-known ability to produce distinctive odors in individuals ingesting it. The extent to which other dietary components alter the sensory characteristics of human milk remains to be investigated. It might be predicted, however, that other foods containing volatile sulfur compounds (eg, onions, broccoli) are likely candidates. Bland foods, eaten by the mothers during both testing days, did not alter milk odors; no change in the odor of the milk was evidenced in collections obtained after lunch compared with those obtained before lunch on the placebo day. However, subtler influences, perhaps evident only if the milk had been swallowed, cannot be excluded.

It is well known that many drugs ingested by a lactating female pass readily into her milk,[17] but little research has focused on the effects such transmission has on the sensory qualities of human milk or on the behavior of the breast-fed infant. The strong flavors of many oral drugs as well as nicotine, caffeine, and alcohol could be expected to alter the odor and consequently flavor of milk. The sensory implications of alcohol transmission to human milk are now under investigation.

Effect of Maternal Diet on the Nursling's Behavior

Garlic consumption by the mothers altered their infants' suckling behaviors such that infants attached to the breast for longer periods of time and sucked more when the milk smelled like garlic. There was a tendency for infants to ingest more milk as well; the lack of a significant effect may be due to inherent limitations on the total amount of milk available to the infant. Because infants suck more slowly from a full breast while taking in larger amounts of milk compared to the faster, nonnutritive sucking from a less full breast,[33] the increased sucking with little difference in milk intake observed in the present study may be indicative of an increase in nonnutritive sucking.

These changes in suckling behavior were probably a consequence of the flavoring of the milk with garlic. Because the adult panelists were able to detect an odor associated with garlic ingestion, it is likely that the infants were as well. Current evidence suggests that human infants are no less sensitive to odors than are adults.[34,35] Moreover, unlike the adult panelists, infants had the distinct advantage of being able to ingest the milk, thereby obtaining retronasal stimulation during swallowing. It is well known that in adults, food aromatics reach the olfactory epithelium during normal chewing and mastication.[36] Because breast-fed infants are able to coordinate swallowing and sucking with essentially no interruption in breathing,[37] we suggest that the mouth movements made during sucking facilitated the retronasal perception of the garlic volatiles present in the milk.

That the changes in the nursling's behavior were a result of changes in the milk cannot be unequivocally concluded from this study, however, because it is possible that the garlic odors, pervading other aspects of the mother such as her breath or sweat, were responsible. However, preliminary data from our laboratory suggest that garlic ingestion alters the breath odors of lactating females within 1 hour of ingestion. Therefore, changes in breath odors are not sufficient to explain the changes in the nursling's behavior because feeds occurring up to 1.5 hour postingestion were no different on the garlic or placebo days. Nonetheless, a comparison of the response to garlic-flavored milk and milk without garlic flavor in the absence of the mother would resolve this issue as well as speak to the question of whether garlic-flavored milk is consumed in greater amounts than unflavored milk.

Presuming the changes in the sensory qualities of milk caused the increased sucking, what was responsible for this effect? Several hypotheses can be suggested. First, there could be something inherently attractive or stimulating about garlic flavor. Sweet sugars are known to be innately positive, stimulating increased sucking and intake of fluids.[38,39] However, there is no evidence that either the taste or flavor of garlic (or its metabolites) is similarly attractive. In fact, there is no evidence for the existence of inherently attractive odors, although the absence of such evidence may reflect the lack of research rather than the true state of things.[10,34,35]

Second, the effects of prior exposure may have contributed to the enhanced suckling response to garlic-flavored milk. Here we discuss two general alternatives. The mothers' recent diets, and most likely their milk, were bland because each mother ate foods low in sulfur during the 3 days preceding each testing day. Thus, any change in flavor may have acted as a stimulant of ingestion for the infant. This may represent a very early version of sensory-specific satiety,[40] the condition where recently consumed foods lose their palatability and are rejected, while more novel foods increase palatability and stimulate food intake.

A second alternative is that the infant's response was due to prior experience with garlic volatiles. As stated earlier, the food frequency questionnaire revealed that each mother occasionally consumed garlic during pregnancy and lactation. The animal literature shows that exposure to odors in utero[41,42] and very early in life[43,44] often results in a preference for these odors. Recently, it has been reported that both the amniotic fluid and body odor of the human neonate can acquire the odors of a spicy meal ingested by the mother prior to delivery.[45] Because the near-term fetus swallows significant amounts of amniotic fluid[46] and has open airway passages which are bathed in amniotic fluid,[47] it may be exposed prenatally to a unique sensory environment. Thus, the efficacy of garlic odors in stimulating increased suckling behavior may have been a consequence of previous experience with these odors via amniotic fluid and mother's milk.

That the flavor of human milk can be altered by the dietary choices of the lactating female has important implications. First, it might imply that bottle-fed babies, who experience a very constant set of flavors from standard formulas, may be missing significant sensory experiences which, until recent times in human history, were common to all infants. Recent studies indicate that 25% of bottle-fed infants experience a change in formula during the first months of life, usually in response to nonspecific symptoms.[48] Perhaps, as Da Mota[49] recently suggested, "this monotony, which is contrary to the most basic rule in dietary habits, might just prove too much for one in four babies."

Second, breast-fed infants may be learning, via flavor cues present in human milk, about the quality of mother's diet. The evidence that early feeding experience influences later food choice in humans is remarkably meager.[50] Much of this work, however, is correlational in nature and/or conducted with older children. Perhaps very early exposure to distinct food flavors, those occurring through breast milk, has more pervasive consequences. The existence of early critical periods for the development of odor preferences in experimental animals and the requirement for physical stimulation during exposure to the odor,[51] a stimulation similar to that obtained during the close body contact of the human mother-infant dyad during nursing, makes this suggestion more likely. Whether the sensory experiences associated with early milk feedings influence the human infant's willingness to accept certain new foods at weaning or thereafter is an important area for future investigation.

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HALLOWEEN TREATS

As a pediatrician and a tennis player, I have acquired an excess of used tennis balls and an aversion to the candy orgy of Halloween. As my solution to both situations, I offered the choice of a tennis ball or a candy bar to the 180 trick or treaters. Ninety-five of 180 kids (53%) chose the tennis ball! Among the girls 15 of 60 (25%) chose the ball; among the boys 80 of 120 (67%) chose the ball. For most it was an agonizing choice! Next year I'm eliminating the candy option.

Stiehm ER. Halloween treats. *Los Angeles Times.* November 6, 1990

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